

Deep Learning & Architectural Foundations: Complete Master Manual

Chapter 1: Classical Foundations of Computer Vision

1.1 Image Filtering and Convolution

Images are grids of pixel values. Convolution applies a kernel that slides across the image.

$$h[m,n] = \sum_k \sum_l f[k,l] * I[m+k, n+l]$$

1.2 Edge Handling and Padding

Zero padding preserves spatial size. Valid convolution reduces output size. Replication extends borders.

1.3 Box Filter Example

3x3 averaging kernel produces smoothing. Example output mean = 50 for uniform patch.

Chapter 2: Neural Network Fundamentals

2.1 Perceptron Model

$$z = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

Activation functions: sigmoid, tanh, ReLU, leaky ReLU

2.2 Bias Role

Bias shifts decision boundary and prevents constraint through origin.

2.3 Example Computation

$$z = 1 + 3x_1 - 2x_2, x = [-1, 2], z = -6, \text{sigmoid approx } 0.00247$$

Chapter 3: Network Topology

Single layer models are linearly separable only.

XOR problem requires hidden layers.

Deeper networks model complex nonlinear boundaries.

Chapter 4: Optimization

4.1 Loss Functions

Binary Cross Entropy and Mean Squared Error define prediction error.

4.2 Gradient Descent

$$W^* = \operatorname{argmin}_W J(W)$$

Methods: SGD, batch, mini-batch

4.3 Backpropagation

Uses chain rule to compute gradients layer by layer.

4.4 Optimizers

Momentum, RMSProp, Adam improve convergence stability.

Chapter 5: Regularization

Dropout randomly disables neurons during training.

Early stopping halts training when validation loss increases.

Chapter 6: U-Net Architecture

Encoder downsamples feature maps. Decoder upsamples them.

Skip connections concatenate encoder and decoder features.

Final 1x1 convolution maps features to class probabilities.

Chapter 7: Multi-Modal Learning

7.1 Face Recognition

Uses embeddings to represent identity in vector space.

7.2 Speech Processing

STFT converts audio into spectrogram images for CNN processing.

7.3 Sequence to Sequence Models

Encoder compresses sequence. Decoder generates output until EOS token.

7.4 Reinforcement Learning

Maps pixel inputs to action outputs for control tasks.